kafka_replication_protocols (Initial Draft)

Domain: Distributed Systems | Document ID: kafka_replication_protocols_v1
Specification Standard: IEEE Capstone Evaluation Corpus

Executive Overview:

Apache Kafka partition replication, in-sync replicas (ISR), high watermark offset, and leader election under network partitions.

Methodological Verification & System Lineage:

This specification is catalogued in IntentCloud for intent-aware hybrid retrieval.
Text chunks are indexed into 384-dimensional dense semantic vectors and 1,000,003-dimensional sparse feature-hashed token projections.
Candidate pools are unified using scale-invariant Reciprocal Rank Fusion (k=60).
Explainable citation attribution is extracted using cross-encoder cross-attention.